Worksheet 6 - The Derivative Function

MATH 1271

09/24/2026

1 Computing the Derivative Function

Last worksheet we computed the derivative of many functions *at a point*, we can now compute their derivative functions!

1. For $f(x) = x^2$, compute $f'(x)$ and verify that $f'(-1)$ matches the value you calculated in the previous worksheet.

2. For $f(x) = \dfrac{1}{x}$, compute $f'(x)$ and verify that $f'(2)$ matches the value you calculated in the previous worksheet.

3. For $f(x) = \sqrt{x+3}$, compute $f'(x)$ and verify that $f'(1)$ matches the value you calculated in the previous worksheet.

4. For $f(x) = \dfrac{1}{x^2+1}$, compute $f'(x)$ and verify that $f'(1)$ matches the value you calculated in the previous worksheet.

5. For $f(x) = x^3 + x - 1$, compute $f'(x)$ and verify that $f'(1)$ matches the value you calculated in the previous worksheet.

6. For $f(x) = \sin(x)$, use the challenge problem of Homework 5 to compute $f'(x)$ and verify that $f'(0)$ matches the value you calculated in the previous worksheet.

2 Extracting Information Using the Derivative Function

1. Suppose that you throw a ball straight up in the air, and know that it's height as a function of time is described by:

$$s(t) = 64 - 16(t-1)^2. \tag{1}$$

(a) What is the derivative function $s'(t)$?

(b) What is the instantaneous velocity of the ball at time $t = \frac{3}{2}$?

(c) What is the best linear approximation of the position of the ball at time $t = 2$?

(d) At which times is the velocity of the ball positive? At which times is it negative? At which times is it zero?

(e) Using what you know about quadratic functions/parabolas, at which time is the ball at it's highest height? Has this time shown up in previous answers?

2. Suppose that the value of a stock for company over a period of ten days is modeled by the function:

$$V(t) = \frac{x^3}{3} - 4x^2 + 15x + 20 \tag{2}$$

where t is the time in days since some initial point and $V(t)$ is the price per share in dollars per share, for $0 \leq t \leq 10$

(a) Suppose that the CEO of the company announced unfavorable news that cause the price of the stock to stop increase and start decreasing. Determine the time t when this announcement occurred.

(b) Suppose that after this news, the board of the company swiftly took action and helped institute changes that caused the price of the stock to stop decreasing and start increasing again. Determine the number of days since the CEO announced the unfavorable news that it took the board to institute these changes.

(c) Graph the function $V(t)$ using a tool such as Desmos etc. At the times you found for the two previous problems, what do you notice about the behavior of the graph near those points? How does the value of the stock at those points compare to the value of the stock at nearby points in time?